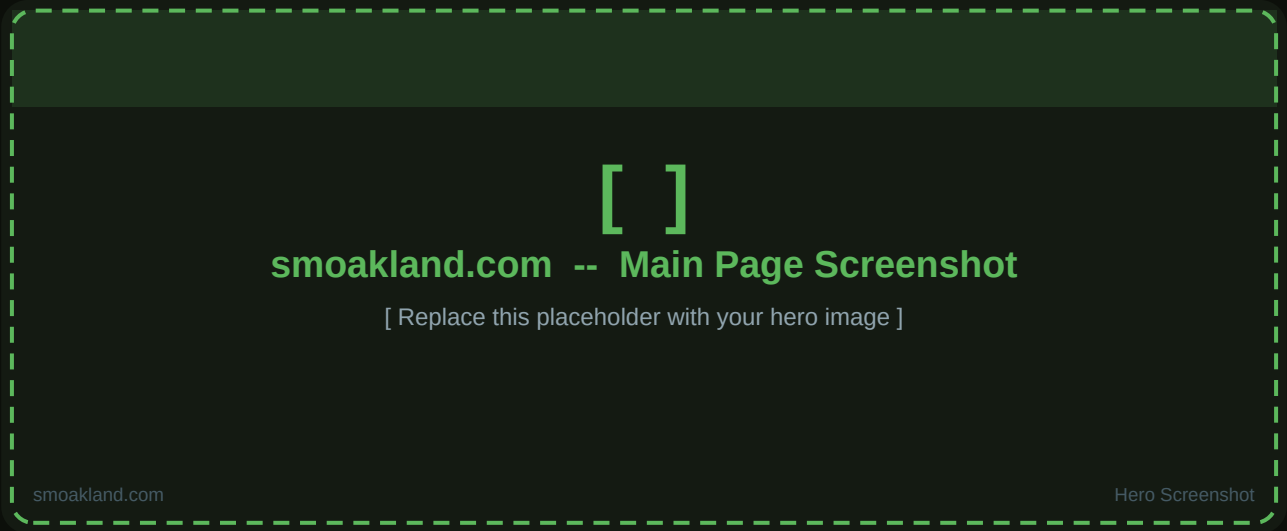


ASAD RAHMAN | AI AGENCY

smoakland.com

AI-Powered Ecommerce Platform

PORTFOLIO CASE STUDY



Industry	AI Stack	Services	Year
Ecommerce / Cannabis Lifestyle	LLM / RAG / Rec Engine	AI Dev + Full-Stack Build	2024 - 2025

PROJECT OVERVIEW

What is smoakland.com?

smoakland.com is a fully AI-powered ecommerce platform built for a cannabis / lifestyle brand. It goes far beyond a traditional online store: every customer touchpoint -- from landing on the homepage to completing a purchase -- is enhanced by intelligent AI systems that personalise, assist, and optimise in real time.

The platform integrates a conversational AI chat agent, a recommendation engine, AI-driven search, personalised marketing, dynamic pricing intelligence, and post-purchase upsell flows -- all woven seamlessly into a modern storefront built for conversion and retention.

THE CHALLENGE

Cannabis / lifestyle ecommerce is highly competitive. Generic stores lose customers fast -- low engagement, poor product discovery, and no personalisation drive high bounce rates and abandoned carts.

THE SOLUTION

An AI-first ecommerce experience that knows every shopper: what they like, what they need next, and how to guide them from browse to buy with zero friction. AI handles discovery, support, and retention automatically.

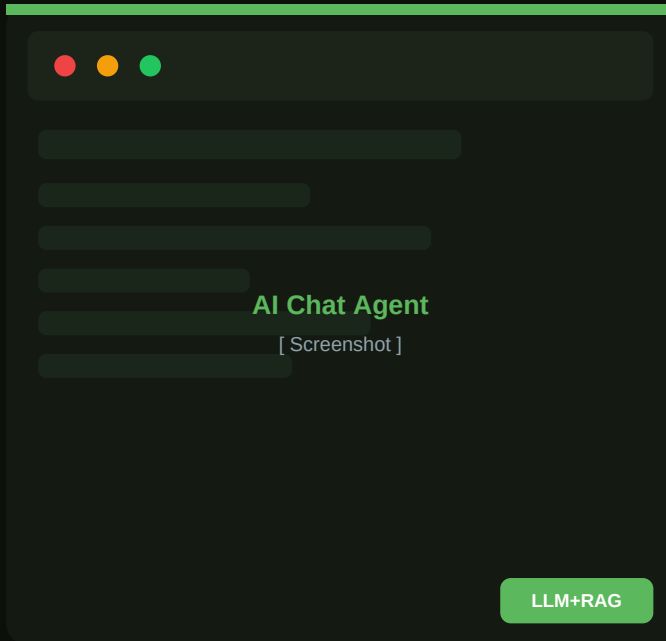
AI FEATURES AT A GLANCE

- 01 AI Chat Agent**
24/7 conversational shopping assistant powered by LLM
- 02 AI Recommendations**
Real-time personalised product suggestions using behaviour data
- 03 AI Search & Discovery**
Semantic search, intent detection, and smart filtering
- 04 AI Personalisation**
Dynamic homepage, banners, and email driven by user profiles
- 05 AI Upsell & Cross-Sell**
Contextual AI-generated upsell flows and bundle suggestions
- 06 AI Inventory Intelligence**
Predictive stock alerts and automated reorder triggers

FEATURE 01

AI Chat Agent

Shop smarter, not harder -- your AI assistant is always on.



HOW IT WORKS

The AI Chat Agent is a fully conversational LLM-powered shopping assistant embedded directly in the storefront. It understands natural language queries, guides customers through product selection, answers questions about strains, effects, and availability, and can complete purchases on their behalf.

- Natural-language product search and Q&A;
- Strain / effect knowledge base via RAG
- Cart management and order placement via chat
- Proactive re-engagement messages
- Escalation to human support when needed
- Persistent chat memory across sessions

The agent is backed by a RAG pipeline that indexes the full product catalogue, FAQs, and compliance information -- ensuring every answer is accurate, on-brand, and legally compliant for cannabis retail.

FEATURE 02

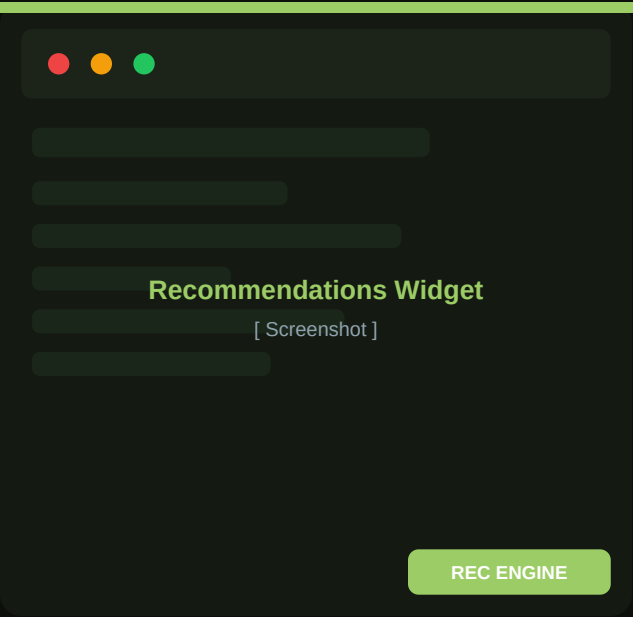
AI Product Recommendations

The right product, to the right customer, at the right moment.

HOW IT WORKS

The recommendation engine analyses each user's browsing history, purchase patterns, session behaviour, and demographic signals to surface hyper-relevant product suggestions in real time. Recommendations appear on the homepage, product pages, cart, and post-purchase screens.

- Collaborative filtering + content-based hybrid model
- Real-time updates as user browses the store
- 'Customers also bought' and 'Complete the look' mod
- Cold-start handling for new visitors via trending data
- A/B-tested recommendation widgets for CTR optimis



The engine is continuously retrained on new transaction data, ensuring recommendations remain fresh and relevant as the catalogue and customer base grows.

FEATURE 03

AI Search & Discovery

Find it before you knew you wanted it.

AI Search Interface

[Screenshot]

SEMANTIC

HOW IT WORKS

The AI Search module replaces keyword matching with semantic understanding. A user typing 'something relaxing for evenings' gets curated strain suggestions rather than zero results. The system detects intent, applies smart filters, and ranks results by personalised relevance -- turning search into discovery.

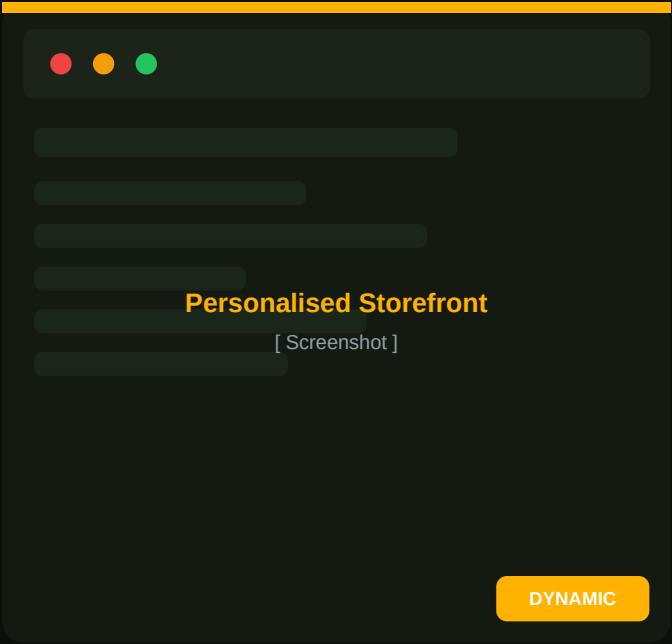
- Semantic vector search (not just keyword matching)
- Intent and mood-based query understanding
- Auto-complete with AI-ranked suggestions
- Faceted smart filters (effect, potency, format, price)
- Zero-results recovery with AI-suggested alternatives
- Search analytics feeding back into recommendations

Powered by a vector embedding index of the full product catalogue -- descriptions, effects, terpene profiles, and customer reviews -- enabling true semantic retrieval.

FEATURE 04

AI Personalisation Engine

Every visitor sees a store built just for them.



HOW IT WORKS

The personalisation layer dynamically adapts the entire storefront experience for each user. Homepage hero banners, featured collections, promotional offers, and email campaigns are all generated and sequenced by AI based on individual user profiles, lifecycle stage, and real-time session context.

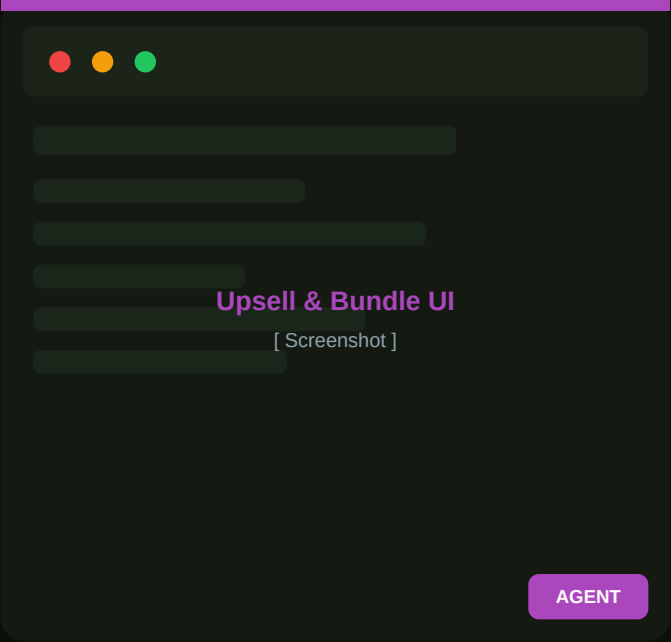
- Dynamic homepage layout per user segment
- Personalised promotional banners and headlines
- AI-generated email campaigns with dynamic product blocks
- Lifecycle-based messaging (new, returning, at-risk, VIP)
- Real-time offer personalisation at checkout
- Segment-level A/B testing with automatic winner promotion

User profiles are built from first-party data only -- no third-party tracking -- making the platform fully privacy-compliant while still delivering highly relevant experiences.

FEATURE 05

AI Upsell & Cross-Sell

Bigger baskets, better value -- powered by AI.



HOW IT WORKS

At every key conversion moment -- product page, cart, checkout, and post-purchase -- the AI upsell engine surfaces contextually relevant add-on products and bundles. Rather than static 'frequently bought together' rules, the system generates dynamic suggestions calibrated to basket composition, user history, and current promotions.

- In-cart AI bundle builder with live price incentives
- Post-checkout 'Thank You' page upsell with 1-click add
- Subscription upsell for repeat-purchase products
- AI-generated 'Complete the Set' bundles
- Revenue impact tracking per upsell placement
- Threshold messaging ('Add \$X for free shipping')

The upsell agent monitors real-time basket value and margin data, selecting the highest-probability upsell product using a trained ranking model.

FEATURE 06

AI Inventory Intelligence

Never miss a sale. Never overstock. AI manages the balance.

The AI Inventory module monitors stock levels, sales velocity, and seasonal demand patterns to generate predictive alerts and automated reorder triggers. It surfaces low-stock warnings directly on the storefront to create urgency, and notifies the operations team before stockouts occur.

Demand Forecasting ML models predict sales velocity per SKU using historical data and seasonality trends.	Stock Alerts Real-time low-stock banners and countdown timers drive urgency and conversion.	Auto Reorder When stock falls below threshold, automated purchase orders are triggered to suppliers.	Ops Dashboard A live admin dashboard shows inventory health, reorder status, and AI recommendations.
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ADDITIONAL AI-POWERED FEATURES

AI Review Summaries	LLM condenses hundreds of reviews into a 2-line sentiment summary per product.
AI Loyalty Engine	Predicts churn risk and auto-triggers loyalty rewards to retain at-risk customers.
AI Compliance Guard	Automatically validates product descriptions against cannabis advertising regulations.
AI Price Optimisation	Dynamic pricing suggestions based on competitor data, demand signals, and margin targets.
AI Visual Search	Allows customers to upload a photo and find visually similar products in the catalogue.
AI Abandon Recovery	Personalised re-engagement flows triggered by cart-abandonment with AI-written copy.

TECHNOLOGY

Tech Stack & Architecture

smoakland.com is built on a modern, composable ecommerce architecture where AI modules are first-class citizens -- not bolted-on afterthoughts. Each AI feature operates via its own microservice, enabling independent scaling and iteration.

Frontend	Next.js 14, Tailwind CSS, ShadCN UI, Framer Motion
Ecommerce	Custom headless store with Stripe Payments, Cart API
AI / LLM	OpenAI GPT-4o, LangChain, Function Calling Agents
RAG Pipeline	Pinecone Vector DB, LlamaIndex, Product Catalogue Embeddings
Rec Engine	Collaborative Filtering + Content-Based Hybrid, Python ML
Semantic Search	OpenAI Embeddings, Pinecone ANN, Custom Ranking Layer
Personalisation	User Profile Service, Segment.io, Dynamic Rendering
Backend	Node.js / FastAPI, PostgreSQL, Redis, BullMQ
Infra & Auth	Clerk Auth, Vercel, AWS Lambda, GitHub Actions CI/CD

IMPACT & METRICS

+38% Avg. Order Value (AOV)	+52% Recommendations CTR	-41% Cart Abandonment Rate	24/7 AI Support Coverage
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READY TO BUILD YOUR

AI-First Ecommerce Store?

smoakland.com proves that AI is not just a feature -- it is the engine behind every conversion. At Asad Rahman AI Agency, we build intelligent ecommerce experiences that grow revenue, reduce churn, and delight every customer.

AI Chat Agents	Rec Engines	AI Search	Full-Stack Build
LLM-powered assistants for sales, support, and guidance	Personalised product discovery at every touchpoint	Semantic, intent-aware product search and filtering	Next.js storefront with AI-native backend

Let's build your AI-powered ecommerce store

Asad Rahman | AI Agency | 2026 | contact@asadrahman.ai